Module 1: What AI Changes

And what it does not change

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COMP 331: AI-Assisted Software Engineering

Primary sources: Osmani, Chapter 1; Thomas & Hunt, Chapters 1-2 (Topics 1-15).

Learning Outcomes

- Explain the difference between vibe coding and AI-assisted software engineering.
- Describe programming with intent.
- Summarize strengths and limits of AI-generated code.
- Interpret Software 2.0 as a shift in specification and verification.

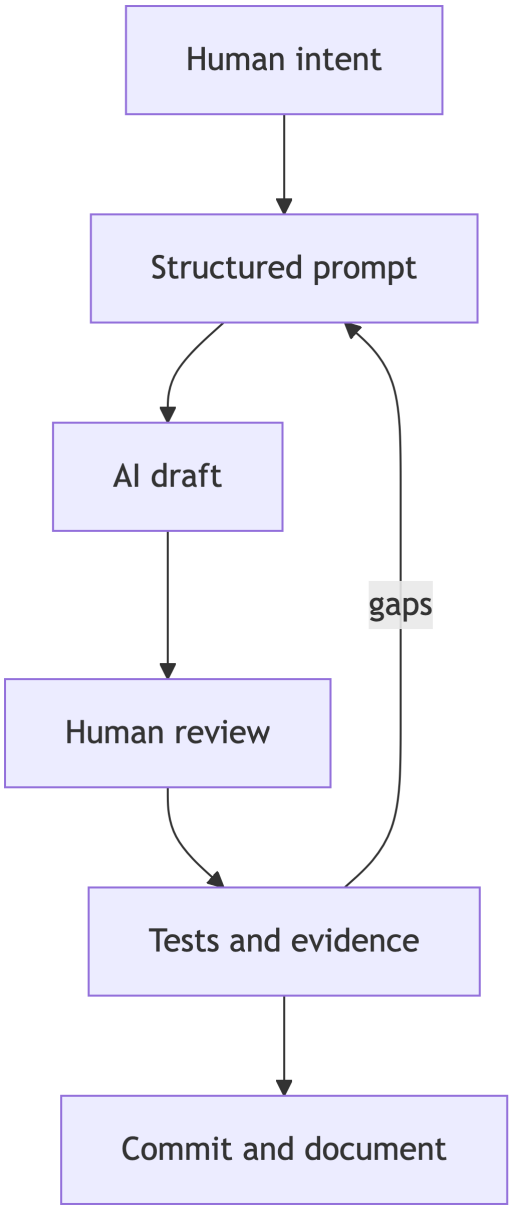
The Core Shift

AI changes the interface to software creation.

It does not remove the need for engineering judgment.

- Before: translate intent into code directly.
- Now: translate intent into prompts, examples, constraints, tests, reviews, and code.
- The human role moves upstream and downstream: specify better, verify harder.

AI-Assisted SWE Loop



Vibe Coding vs Engineering

Vibe Coding	AI-Assisted Engineering
Conversational exploration	Structured collaboration
Accepts plausible output	Verifies behavior and risks
Optimizes for momentum	Optimizes for trusted progress
Weak artifact trail	Prompts, tests, decisions, commits

Programming With Intent

Programming with intent means making the desired behavior explicit before accepting implementation.

- What should the system do?
- What should it refuse to do?
- What edge cases matter?
- What evidence proves it works?
- What human decision remains non-delegable?

Pragmatic Foundation

Thomas and Hunt start with a philosophy: pragmatic programmers take responsibility, keep learning, communicate clearly, and adapt to change.

For AI-assisted SWE, this means:

- Own every generated artifact.
- Communicate precisely with humans and tools.
- Treat AI output as a draft, not authority.
- Keep improving your knowledge portfolio.

Pragmatic Approach

The second reading turns philosophy into design habits.

- Good design is easier to change.
- DRY: keep each piece of knowledge in one authoritative place.
- Orthogonality: avoid coupling unrelated decisions.
- Reversibility: avoid choices that trap the project.
- Tracer bullets and prototypes help test understanding early.

AI Does Not Replace Design Judgment

AI can generate code quickly, but it cannot know which future changes matter unless we supply context.

Pragmatic questions still apply:

- Is this easier to change?
- Did the AI duplicate knowledge or intent?
- Are components coupled for no good reason?
- Is this a reversible experiment or a commitment?
- What estimate or prototype would reduce uncertainty?

Where AI Helps

- Drafting boilerplate and examples.
- Exploring alternate implementations.
- Explaining unfamiliar APIs.
- Creating first-pass tests and docs.
- Speeding up setup and debugging paths.

Where AI Still Struggles

- Hidden requirements and domain context.
- Architecture tradeoffs across time.
- Security-sensitive boundary decisions.
- Correctness when tests are weak.
- Knowing when the task is underspecified.

Software 2.0 Lens

Software 2.0 reframes some development as training, prompting, selecting, and evaluating behavior.

In this course, we apply that lens pragmatically:

- Prompts become one specification artifact alongside requirements, tests, and code.
- Code review includes model-output review.
- Engineering evidence matters more than confidence.

Initial Setup: Establishing the Workflow

Assignment 1 is not just tool setup.

It establishes the course habit:

- Capture the prompt.
- Inspect the suggestion.
- Make a manual decision.
- Verify the result.
- Document what happened.

Practice Activity

Ask Copilot for a small Java method.

Then improve the interaction:

- Add input constraints.
- Add expected behavior.
- Add one edge case.
- Ask for a JUnit test.
- Record what changed between attempts.

Takeaways

- AI assistance amplifies both good and bad engineering habits.
- Intent, verification, and responsibility remain human work.
- The course evidence trail is part of the engineering process.